

UBT411: ANIMAL AND PLANT BIOTECHNOLOGY LAB

L T P Cr

0 0 4 2.0

Course Objectives: The course will enable the students to design and undertake various experiments in genetic and metabolic engineering, animal biotechnology, and plant biotechnology

Detail contents:

Laboratory design and layout for animal and plant tissue culture, To check for bacterial and fungal contamination in the tissue culture laboratory, Preparation of animal and plant tissue culture media, Isolation of mononuclear cells and cell counting and viability, To carry out surface disinfection and establishment of aseptic cultures of plant tissue, Lymphocyte cell culture, To study the effect of plant growth regulators on morphogenesis, To test the kanamycin resistance in plant tissue to identify selection marker for genetic transformation, Preparation of competent cells and genetic transformation of *Agrobacterium tumefaciens* using the freeze-thaw method, To perform the histochemical assay for the analysis of putative transformed tissue, To confirm the stable genetic transformation by amplification of the selection marker gene, To design primers for the amplification of a given gene sequence, To test the clonal fidelity of micropropagated plants using RAPD markers, SNP genotyping by PCR-RFLP method.

Course Learning Outcomes (CLO):

Students will be able to

1. design the animal and plant tissue culture labs
2. prepare media for plant and animal tissue culture
3. induce various morphogenetic events in plant tissue culture
4. undertake plant genetic transformation and genotyping

Text Books:

1. Experiments in Plant Tissue Culture By Johan H Dodds and Lorin W Roberts. Publishers: Press Syndicate of University of Cambridge, 1990
2. Freshney's Culture of Animal Cells: A Manual of Basic Technique and Specialized Applications, 8th Edition 2010

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

3. Molecular cloning. A laboratory manual by T Maniatis, E F Fritsch and J Sambrook. pp 545. Cold Spring Harbor Laboratory, New York

Reference Book:

1. Masters, J. R.W., Animal Cell Culture, Oxford (2000) 3rd edition
2. Razdan, M.K., Introduction to Plant Tissue Culture, Science Publishers (2003).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	Sessionals (May include assignments / quizzes / performance)	100

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